

Graduation Project Conference Report

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Pick at least two posters that were interesting. (1) Summarize the methods and findings of the poster, (2) discuss the impact or implications of the result, and (3) take a photo of the poster

Poster pick 1

- Summary

Wearable devices can detect acute stress using physiological signals, but most stress detection models are trained on a single dataset and often fail to generalize to new users, devices, or environments due to domain shift. This study investigates whether a wearable stress detection model can achieve robust cross-dataset performance through subject-wise normalization. By normalizing each individual's physiological features to their own baseline, we reduce person-specific and device-specific variations and obtain more domain-invariant representations. Experiments were conducted on two independent Empatica E4 datasets, WESAD and Exercise-Stress, using a shared 52-feature extraction pipeline and subject-independent evaluation. Results show consistent improvements in both within-dataset and cross-dataset settings, with the largest gains observed in transfer scenarios, demonstrating that baseline offset is a major source of domain shift. Furthermore, the stress predictions are integrated into an LLM-based support agent that provides personalized responses using retrieval-augmented generation (RAG). The proposed approach offers a simple, label-free method for improving the generalizability of wearable stress detection systems.

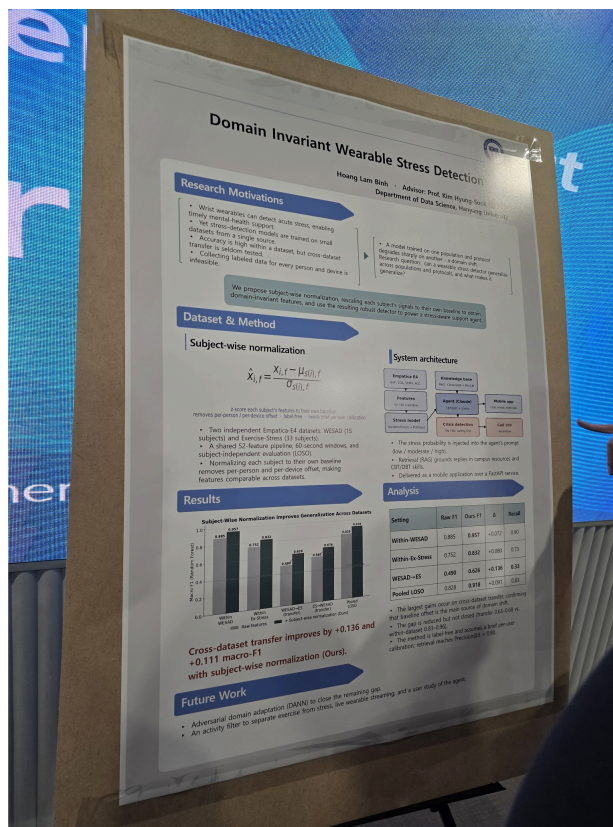
- Implications

The results demonstrate that subject-wise normalization is an effective and practical approach for improving the generalizability of wearable stress detection models. The substantial performance gains in cross-dataset transfer experiments suggest that individual baseline differences are a major source of domain shift in physiological sensing data. By reducing these differences, a model trained on one population can be applied more reliably to new users and environments without requiring extensive labeled data collection.

These findings have important implications for real-world deployment of wearable mental health applications. Since collecting and labeling stress data for every user is costly and often impractical, a label-free normalization approach provides a scalable alternative for building robust stress monitoring systems. The improved generalization also enables wearable devices to support a broader range of users, devices, and usage scenarios.

Furthermore, integrating the stress detector with an LLM-based support agent demonstrates the potential of combining physiological sensing with conversational AI. Such systems could provide timely stress-aware interventions, mental health resources, and personalized support in everyday settings. Future work on domain adaptation techniques and real-time deployment may further enhance reliability and facilitate wider adoption of wearable stress monitoring technologies.

- Photo



Poster pick 2

- Summary

This study presents a data-driven decision-support system for a District Heating (DH) network that combines classical forecasting techniques with a vision-language-based anomaly detector. The central thesis is that classical forecasting and pretrained multimodal models are complementary rather than competing, and that trustworthy fault detection arises from corroboration across independent detectors (statistical, visual, and physics-based) rather than from any single model.

The dataset consists of proprietary records from the PNS-8 station of a real Russian DH utility, comprising 32,547 hourly observations from 2021 to 2024 (32,530 after cleaning) across nine variables. A defining challenge of the dataset is the complete absence of anomaly labels, which motivates much of the methodological design in the anomaly-detection track.

The system is organized into two tracks. Track A (Forecasting) targets the return temperature and adopts a Direct multi-output approach over a recursive one, because the direct method avoids error accumulation and remains robust to anomalous inputs. The final model is a Direct LightGBM tuned with Hyperopt (TPE) and evaluated with 24-hour rolling-refit backtesting, achieving an smMAPE of 8.31%.

Track B (Anomaly Detection) addresses the missing-label problem by generating statistical pseudo-labels (Z-score, IQR, rate-of-change) and applying a VLM4TS pipeline built on a frozen CLIP ViT-B/16 screen. The pipeline incorporates multi-period STL deseasonalization, overlapping six-month slices, and three-channel agreement, with a vision-language model serving as an arbiter over the ambiguous intervals. As a result, recall against the pseudo-labels improved from 27 to 80.5%, and the detected anomaly intervals expanded from 11 to 108, organized into four confidence tiers. This structure transforms the output from simple detection into a triaged, prioritized fault assessment.

- Implications

The multi-detector corroboration design escalates alerts only when independent detectors agree. This allows operators to prioritize high-confidence anomaly intervals and respond effectively without being overwhelmed by false alarms, making the system practically actionable for aging DH infrastructure.

The study further demonstrates that pretrained multimodal models can be repurposed as novel verification and arbitration tools for unlabeled industrial time series. Rather than learning the time series from scratch, the VLM adds value precisely in the ambiguous boundary cases where statistical and physics-based detectors are inconclusive.

At the same time, several limitations should be acknowledged. The statistical pseudo-labels are considerably denser ($\approx 10.5\%$) than the sparse-regime assumption ($\approx 1\text{--}3\%$) underlying VLM4TS, and there is no expert-annotated ground truth against which true precision and recall can be measured. Moreover, the analysis relies on data from a single pumping station, so cross-network generalization remains untested. These constraints temper the reported gains and point toward future work, including the acquisition of expert-annotated anomalies and validation across multiple stations.

- Photo

